1 End-to-end observability

This chapter covers

- What we mean by observability
- Why observability matters
- An end-to-end example of observability
- Challenges of cloud-native systems and how observability can help

In cloud-native environments, such as public cloud offerings like AWS or on-premises infrastructure (e.g., a Kubernetes cluster), one typically deals with many moving parts. These parts range from the infrastructure layer, including compute (e.g., VMs or containers) and databases, to the application code you own.

Depending on your role and the environment, you may be responsible for any number of the pieces in the puzzle. Let's have a look at a concrete example: consider a serverless Kubernetes environment in a cloud provider. In this case, both the Kubernetes control plane and the data plane (the worker nodes) are managed for you, which means you can focus on your application code in terms of operations.

No matter what part you're responsible for, you want to know what's going on so that you can react to and, ideally, even proactively manage situations such as a